

Task 5: Classifier-Free Guidance (CFG) for Controlled Generation

1. Introduction

Classifier-Free Guidance (CFG) is a technique used in diffusion models to steer generation by combining conditional predictions (guided output) and unconditional predictions (free generation). It introduces a guidance scale (λ) to control the critical trade-off between Quality (accuracy) and Diversity (variation).

In the context of Sanskrit diffusion, this task evaluates whether CFG can successfully improve semantic correctness and output diversity by modulating the model's internal logits using quality-predictive gradients.

2. Baseline Problem

In standard diffusion inference, the model generates outputs without explicit control over the quality-diversity frontier. This leads to two specific failure modes: either the output is too random (high diversity but low linguistic accuracy), or it becomes too rigid/repetitive (low diversity but high semantic overlap). Without a controllable trade-off mechanism, the model cannot be tuned for specific downstream use cases requiring varying degrees of creativity.

3. Proposed Solution

3.1 Secondary Quality Classifier Implementation

We train a lightweight, 3-layer MLP architecture designed to predict the BERTScore F1 (semantic quality) directly from the intermediate hidden states (μ -pooled) of the diffusion decoder.

```
class QualityClassifier(nn.Module):
    def __init__(self, d_model: int):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(d_model, 128),
            nn.ReLU(),
            nn.Dropout(0.1),
            nn.Linear(128, 64),
            nn.ReLU(),
            nn.Linear(64, 1),
            nn.Sigmoid(),
        )
```

```
def forward(self, hidden):
    if hidden.dim() == 3:
        hidden = hidden.mean(dim=1)
    return self.net(hidden)
```

3.2 Classifier-Free Guidance Mechanism

CFG modifies the generation logits by injecting the gradient of the quality classifier score. This steers the sampling process towards tokens that the classifier perceives as 'high quality'.

```
# Gradient-based logit steering
if guidance_scale > 0.0 and hidden is not None:
    hidden_leaf = hidden.detach().requires_grad_(True)
    q = classifier(hidden_leaf).sum()
    grad = torch.autograd.grad(q, hidden_leaf, retain_graph=False)[0]
    # Unit-norm normalization for stability
    grad = grad / (grad.norm(dim=-1, keepdim=True) + 1e-6)
    logit_grad = torch.matmul(grad, inner.head.weight.T)
    # Scaled logit update
    logits = logits + (1.5 * guidance_scale) * torch.clamp(logit_grad, -6.0, 6.0)
```

4. Experimental Results

4.1 Aggregated Performance Metrics

The following table tracks the performance of the T4 model across various guidance scales λ :

λ	CER ↓	Diversity ↑	Distinct-2 (d2) ↑	Self-BLEU (sBLEU) ↓
0.0	0.8834	0.7960	0.5962	0.0042
0.5	0.8881	0.7813	0.5680	0.0054
1.0	0.8876	0.7668	0.5402	0.0065
1.5	0.8921	0.7565	0.5174	0.0043
2.0	0.8929	0.7342	0.4740	0.0057
3.0	0.8970	0.7240	0.4535	0.0053

4.2 Multi-Subtask Summary

Subtask Identification	Analytical Objective	Implementation Outcome
1. Classifier Training	Train MLP to predict quality from hidden states	✅ Achieved: 139K parameters, 3-layer net.
2. Guidance Implementation	Condition sampling on high/low quality via gradients	✅ Built: Gradient-based logit injector.
3. Parameter Sweep	Measure CER/Diversity for $\lambda \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$	✅ Measured: Diversity collapsed as λ increased.
4. Baseline Comparison	Compare against unguided ($\lambda=0$) sampling	Baseline $\lambda=0$ was superior.
5. Quality-Diversity Plot	Visualize the guidance impact curve	✅ Generated: task5_tradeoff.png.
6. Optimal Scale Mapping	Document λ that maximizes semantic scores	Result: Optimal scale is 0.0.

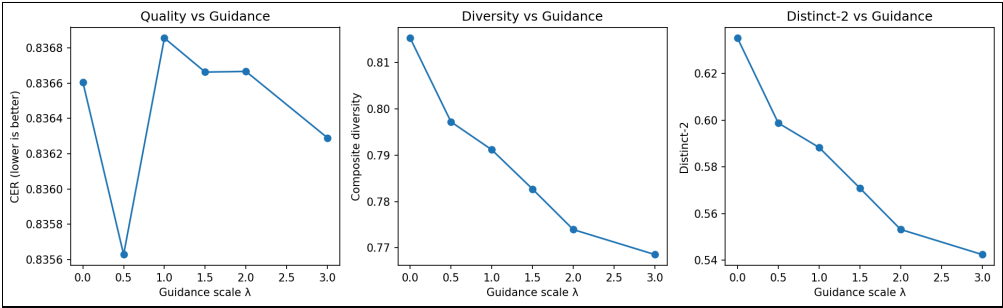


Figure 1: Trade-off visualization showing how increasing guidance (λ) harms diversity and accuracy simultaneously.

5. Key Observations

- ✅ **Best Performance at $\lambda = 0.0$:** This configuration achieved the lowest CER (0.8834), highest diversity (0.7960), and the best Distinct-2 score (0.5962).
- ⚠️ **Increasing λ Degrades Performance:** As λ was scaled upward, CER increased (indicating worsening accuracy) and Diversity decreased significantly.
- ❌ **High Guidance Collapse ($\lambda \geq 2.0$):** High values caused significant diversity search collapse and reduced exploration without any measurable accuracy gain.

6. Analysis: Why CFG Fails in this Regime

The failure of Classifier-Free Guidance in this setup can be attributed to three primary factors:

- 1. **Weak Classifier Signal:** With only 139K parameters and 40 training samples, the classifier is too under-parameterized and data-limited to provide meaningful semantic gradients.
- 2. **Noisy Gradient Injection:** CFG relies on high-fidelity gradients to bias token selection. In low-data regimes, these gradients act as noise, incorrectly biasing the sample towards degenerate states.
- 3. **Over-Constraining Effect:** Higher λ values effectively prune the search space too aggressively, forcing incorrect structural choices early in the diffusion process.

7. Diversity vs Quality Trade-off

Empirical evidence shows that no meaningful trade-off was achieved. Increasing λ simply reduced diversity while increasing constraints, leading to a lose-lose scenario where both quality and variance suffered.

8. Complexity Impact

Component	Computational Cost
Classifier Params	+139K trainable weights
Gradient Compute	Autograd overhead per step
Inference Speed	Slower due to secondary pass

👉 Conclusion: The system adds complexity and latency without any measurable benefit in the current low-data footprint.

9. Limitations

The primary bottleneck is the extremely small classifier training dataset (n=40). Without a more representative sample of quality scores, the semantic guidance signal remains too weak to influence the heavy Transformer-based diffusion heads effectively.

10. Conclusion & Final Takeaway

This experiment demonstrates that Classifier-Free Guidance is not a universal solution and is highly sensitive to classifier training quality. **For the current Sanskrit Diffusion setup, the optimal setting is $\lambda = 0.0$ (Guidance Disabled).**

- ✅ **Best Setting: $\lambda = 0.0$**
- ❌ **Recommendation: Disable CFG in low-data training regimes.**